RAJ MAHESH SHARMA

+91 9359674711 ◇ Sillod, Maharashtra, India
r.m.sharma.1069@gmail.com ◇ LinkedIn ◇ GitHub ◇ **Portfolio**

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science, Yeshwantrao Chavan College — Expected 05/2027
Relevant Coursework: Data Structures & Algorithms, SQL, OOP, Math, Statistics — CGPA: 8.44/10

SKILLS

Programming & Databases	Python, C, TypeScript, JavaScript, SQL (MySQL)
AI & Core Competencies	Multi-Agent Orchestration, RAG, Predictive Modeling, Data Analytics
Frameworks & Libraries	FastAPI, LangGraph, OpenCV, PyTorch, Pandas, Numpy, Scikit-Learn
Tools & Infrastructure	Docker, ChromaDB, Git, GitHub, Jupyter Notebook

PROJECTS

Dynamic Multi-Agent Orchestrator | GitHub — *05/2026 - Present*
Python, LangGraph, FastAPI, Docker, SQL, Server-Sent Events (SSE)

- Built a multi-agent system with FastAPI and LangGraph that routes tasks across 4 specialized agents for decomposition, retrieval, critique, and synthesis.
- Added Python and SQL tools with explicit error handling and streamed responses in real time using Server-Sent Events.
- Added context-window pruning for long reasoning chains and built an evaluation loop to test and improve agent responses.
- Exposed the system through a 5-endpoint REST API with structured logging and background workers, and containerized the application with Docker Compose.

VidQuery: Multimodal AI Video Search Engine | GitHub — *03/2026 - Present*
Python, OpenCV, YOLOv8, CLIP, ChromaDB, EasyOCR, PyTorch, SciPy

- Built a video search pipeline using semantic text queries and object detection with YOLOv8.
- Created a 1 FPS frame-extraction system using CLIP (ViT-B-32) and ChromaDB, reducing average retrieval latency to <1 second across **[10,000+]** indexed frames.
- Implemented cross-frame object tracking using Intersection over Union (IoU) and the Hungarian algorithm.
- Added EasyOCR and K-Means clustering to index text and dominant colors directly from video frames.

AWARDS & CERTIFICATIONS

Naukri Campus Young Turks Contest — Data & AI Track — *2025*
Top 3% National Rank

Oracle Cloud Infrastructure 2025 AI Foundations Associate — *September 2025*
Issued by Oracle · Validated expertise in Cloud and AI fundamentals, deploying ML workflows on OCI, and Responsible AI.

LEADERSHIP EXPERIENCE

Lead Coordinator (Campus Ambassador) — *07/2025 - Present*
Career Katta Initiative (MITSC)

- Led a 12-member team to execute a state-wide skill development program at the campus level.
- Organized campus-wide campaigns and daily digital workshops for career guidance, driving active participation from over 100 students.
- Served as the primary liaison between college administration and the student body to connect students with industry experts.